ON-OFF
MODES
TAPE
FUNCTIONS
LIGHTS

TURNING MACHINE ON Place switch on left rear to ON. Key: PRIME, CLEAR, Set RUN mode.

MODES OF OPERATION

RUN
Mode for manual calculations and program execution.

LEARN
Used when entering a new program from keyboard into memory. Displays next step.

LEARN and PRINT
Learns and prints each step as entered.

LIST PROGRAM
Paper tape listing of program stored in memory. Key: PRIME GO

Works only on machine with printer

TAPE DRIVE

(on models so equipped)

RELEASE
Unlocks cassette.

FORWARD
Hold down to wind tape in forward direction.

TAPE READY
Locks head in contact with tape.

REWIND
Hold down to rewind tape.

LOCK KEYS

FL / SC
Down for conversion to Scientific Notation.

RADIANS / DEGREE
Up for radians input. Down for degree input. (See conversion key on function section)

PAPER FEED
Advances paper one line when depressed. (See program code.)

PRINTER ON
Down to turn Printer ON.

ERROR LIGHTS

Prog Error Flashes when illegal operation occurs; also, when overflow or wrong program address occurs (see error test).

Mach Error Flashes when program is not loaded properly from tape cassettes. To correct: PRIME, repeat operation.

OPERATION SELECTOR KEYS

These keys select operations in storage registers 00 thru 15 **EXAMPLE:** 2 X 3 = 6 performed in Reg. 00. Depress 2 ST REG 00 3 X REG 00.

Sets registers 00 thru 15 to zero. (except display)

CLEAR 0014 | **T** 01 | **+** 02 | **×** 03 | **÷** 04 | **T** 05 | **ST** 06 | **RE** 07 | **f(x)** 10 | **SP** 08

Selects register keys 00 thru 15 to be subroutines. (One step from keyboard or program.)

FUNCTION KEYS

PRIME Sets display to zero. Sets prog. ctr. to zero. Turns error light off.	SIN^{-1} **SIN**	COS^{-1} **COS**
DEG-RAD **RAD-DEG**	TAN^{-1} **TAN**	**RETURN** is used at the end of a programmed subroutine. **1/X**
SHIFT is required for UPPER CASE functions. Light indicates shift mode. e.g. $LOG_{10}2$ = Key: 2 SHIFT $LOG_{10}X$ Read: .3010299956	$LOG_{10}X$ LOG_eX	10^X e^X
	INT X X^2	**|X|** $\sqrt{X}$

LEFT/RIGHT
REGISTER
PROG. KEYS
PRINTER
NON-PROG. KEYS

LEFT REGISTER (uses Reg. 15, right register is identical, uses Reg. 14)

T displays and clears register.

MINUS subtracts number in display from number in reg. Difference is displayed.

÷= divides number in reg by number in display. Quotient displayed

PLUS adds number in display to number in register and displays sum.

×= multiplies number in display. Product displayed.

STORE stores displayed number in register.

RECALL recalls number from register to display.

EXAMPLE:

$$\left\{\frac{(2 \times 3) - 4}{5}\right\} + 6 =$$

Key: 2 Store 3 ×= 4 – 5 ÷= 6 +

Read: 6.400000000

CHANGE SIGN Changes sign of number or exponent in display.

CLEAR DISP. Clears display.

SET EXP Used for setting exponent. See example.

EXAMPLE:

-123.456×10^{-78}

Key:

CHANGE SIGN

1 2 3 . 4 5 6

SET EXP

CHANGE SIGN

7 8

STORE

Read:

-1.234560000-76

PROGRAMMING KEYS

END = Last step in program.

LOAD PROGRAM transfers program from tape to memory.

MARK followed by any code sets flag at step.

SEARCH followed by any code branches uncond. To corresponding MARK flag.

STOP As a program step, it stops program execution.

GO starts or continues program after stop.

DECISIONS

≠0 jumps 2 program steps if display is not zero.

J if 0 Jumps 2 program steps if display is zero.

ERROR jumps 2 program steps if illegal operation, address or overflow. (See Error Lights.)

J if + Jumps 2 program steps if display is positive.

EXTRA MEMORY **(1 register uses 8 program steps – 2-step Command)**
(Operation Selector Key must be in T, + or – position; max. 40 Register additional)

STORE followed by any register Key 00 thru 15, stores displayed number into indicated register.

RECALL followed by any register Key 00 thru 15, recalls number into display.

MANUAL PRINTER CONTROL

α

PRINT 1-step manual operation from keyboard.

EXAMPLE:
To Set Print Condition
Key: SET PC 0 0 3
(PRINT will print 3 decimal places. Unless Prime is depressed or program is run, the Print condition remains the same.)

To Print:
Key: 1 2 3 . 4 5 6 7 8 9 PRINT
Prints: 123.456 K

Indicates keyboard control

Last Digit of P.C.	Output Format
0	Floating exponential
1	2 places
2	3
3	4
4	5
5	6
6	7
7	8
8	9 places
9	

Corrections:
- Manual PRINT does not print a "K".
- The (later?) 600 models require 4-digit entry for SET PC regardless of memory size.
- The max additional registers varies by by model:
  - 600-2: 39
  - 600-6: 103
  - 600-14: 231

NON-PROGRAMMABLE KEYS

SET P.C. Sets program counter to step number with next 3 keystrokes. Also sets Print condition.

VERIFY PROG. Sums all program codes from Step 000 to END.

RECORD PROG. Transfers program from memory to tape cassette.

STEP Interrupts program execution, in RUN-mode. Allows single step execution. Advances step number in LEARN-mode for visual checking of program.

PROGRAM CODES

CODE	KEY
0000	0
0001	1
0002	2
0003	3
0004	4
0005	5
0006	6
0007	7
0008	8
0009	9
0010	.
0011	SET EXP
0012	CHS
0013	SEARCH & STOP
0014	CLEAR
0015	CLD
0114	TOTAL R
0115	TOTAL L
0214	+ R
0215	+ L

CODE	KEY
0314	– R
0315	– L
0414	× R
0415	× L
0514	÷ R
0515	÷ L
0614	ST R
0615	ST L
0714	RE R
0715	RE L
0800	SEARCH
0801	RECALL
0802	PRINT
0803	GO
0804	J if 0
0805	J if +
0806	SIN
0807	COS
0808	TAN
0809	RAD – DEG
0810	$Log_e X$

CODE	KEY
0811	e^X
0812	X^2
0813	√X
0814	LOAD PROG
0815	1/X
0900	MARK
0901	STORE
0902	α
0903	STOP
0904	J if ≠ 0
0905	J if ERROR
0906	SIN^{-1}
0907	COS^{-1}
0908	TAN^{-1}
0909	DEG – RAD
0910	$Log_{10} X$
0911	10^X
0912	INT X
0913	|X|
0914	END
0915	RETURN

SPECIAL REGISTERS

CODE	KEY
XX00	00
XX01	01
XX02	02
XX03	03
XX04	04
XX05	05
XX06	06
XX07	07
XX08	08
XX09	09
XX10	10
XX11	11
XX12	12
XX13	13
XX14	14
XX15	15

NOTE: XX indicates the register to perform selected operation. (See Operation Selection.)

OPERATION SELECTION

CODE	SWITCH
01YY	TOTAL
02YY	+
03YY	–
04YY	×
05YY	÷
06YY	STORE
07YY	RECALL
08YY	SP
10YY	f(x)
14YY	SP & ST *

NOTE: YY is one of the registers from 00 thru 15. (See Special Registers.)
* Swap Reg. YY with display.

PROGRAM PRINTER CONTROL
(preceded by PRINT)

DECIMAL CONTROL

CODE	KEY	NO. OF DECIMALS
XX00	00	0
XX01	01	1
XX02	02	2
XX03	03	3
XX04	04	4
XX05	05	5
XX06	06	6
XX07	07	7
XX08	08	8
XX09	09	9
XX10	10	FLOATING
XX11	11	EXPONENTIAL
0015	CLD	PAPER FEED

NOTE: XX indicates the letter to be printed (See Alpha Codes)

ALPHA CODES

OPERATION KEY SETTING		CODE	LETTER
SP up	all up	00YY	X
SP	T down	01YY	Y
SP	+	02YY	Z
SP	–	03YY	A
SP	×	04YY	B
SP	÷	05YY	C
SP	ST	06YY	D
SP up	RE down	07YY	E
SP down	all up	08YY	F
SP	T down	09YY	G
SP	+	10YY	H
SP	–	11YY	I
SP	×	12YY	J
SP	÷	13YY	K
SP	ST	14YY	L
SP down	RE down	15YY	M

NOTE: YY controls decimal (See Decimal Control)

DECIMAL SHIFT COMMANDS
(preceded by α)

SHIFT RIGHT

CODE	MULTIPLY
1001	10^1
1002	10^2
1003	10^3
1004	10^4
1005	10^5
1006	10^6
1007	10^7
1008	10^8
1009	10^9
1010	10^{10}
1011	10^{11}
1012	10^{12}
1013	10^{13}
1014	10^{14}
1015	10^{15}

SHIFT LEFT

CODE	DIVIDE
1101	10^1
1102	10^2
1103	10^3
1104	10^4
1105	10^5
1106	10^6
1107	10^7
1108	10^8
1109	10^9
1110	10^{10}
1111	10^{11}
1112	10^{12}
1113	10^{13}
1114	10^{14}
1115	10^{15}

MISCELLANEOUS f(x)

CODE	KEY	OPR
0902	α	} π
1000	00	

CODE	KEY	OPR
0902	α	} PAUSE
0903	STOP	

ADDITIONAL CODES Preceded by Code 0902 (α key)

CODE	KEY	COMMAND
0801	RECALL	Loads all data registers from number in display through register 0.
0804	J if 0	Register 0 = Display
0805	J if +	Register 0 ≥ Display
0806	SIN	Register 0 < display
0810 0910	Log_eX or $Log_{10}X$	Trace "ON"
0811 0911	e^X or 10^X	Trace "OFF"
0814	LOAD PROG	Bypasses number of blocks displayed and loads the next block.
0901	STORE	Records all data registers from number in display through register 0.

preceded by Code 1502 (I/O key on 600)

CODE	KEY	COMMAND
0200	00	Register 0 ≡ Display
0300	00	Register 0 ≠ Display

CODE	KEY
1502	I/O
1511	INDIR

CODE	KEY
1513	GROUP 1
1514	GROUP 2

NON PROGRAMMABLE FUNCTIONS

RUN LEARN

When RUN and LEARN keys are locked down simultaneously, the operations of the following keys are changed to the operations in the parentheses.

SET P.C. (INSERT)
(makes space by inserting GO (0803) and advances the following stored program steps by 1 step.)

(BACKSTEP) VERIFY PROG.
(decreases program counter 1 step.)

(SEARCH, MARK, PRIME STOP)
(followed by any code branches to the corresponding MARK flag and stops.)

(DELETE) RECORD PROG.
(erases displayed program step and moves stored program 1 step back.)

Correction: The "alpha numeric comparisons" codes are not 0200 and 0300, they are 1400 and 1500.

INDIRECT ADDRESSING

INDIR.

Allows you to access the memory indirectly and perform arithmetic operations.

EXAMPLE:

Before:

Reg. 05 = 30.

Reg. 30 = 20.3

Display = 3.

Key: INDIR, × 05

After:

Reg. 05 = 30

Reg. 30 = 60.9

Display = 60.9

ADDITIONAL KEYS ON 600 SERIES

I/O
(Input/Output)
a) address peripherals
b) alpha/numeric decisions

GROUP 1
used to address peripherals

GROUP 2
used to address peripherals

AUTOMATIC PRINT MODE
(all models with printer)

To turn Automatic Print ON
Key: SHIFT PRINT PRINT
Display and operation are printed when keyboard is used manually.

To turn Automatic Print OFF
Key: SHIFT PRINT SHIFT PRINT

TRACE MODE

To turn Trace Mode ON
Key: SHIFT PRINT LOG_eX
Prints display and operation symbol of every step during program execution.

To turn Trace Mode OFF
Key: SHIFT PRINT e^X